# SubnettX

Zero-Trust Dual-Vector Attendance & Threat-Monitoring

INITIALIZING SYSTEM PROTOCOLS...

# THE FLAWS IN LEGACY SYSTEMS

### Manual Roll Call

Wastes 10–15 minutes of valuable lecture time and is highly prone to human error and proxy attendance.

### Static QR Codes

Easily bypassed. Students photograph the QR, send it via WhatsApp, and friends scan it from their dorm rooms.

### GPS & Geofencing

Vulnerable to basic spoofing techniques using fake GPS applications or VPN tunneling.

### Bluetooth Beacons

Requires expensive, proprietary hardware installations in every classroom and drains student phone batteries

# ZERO-TRUST DUAL-VECTOR ARCHITECTURE

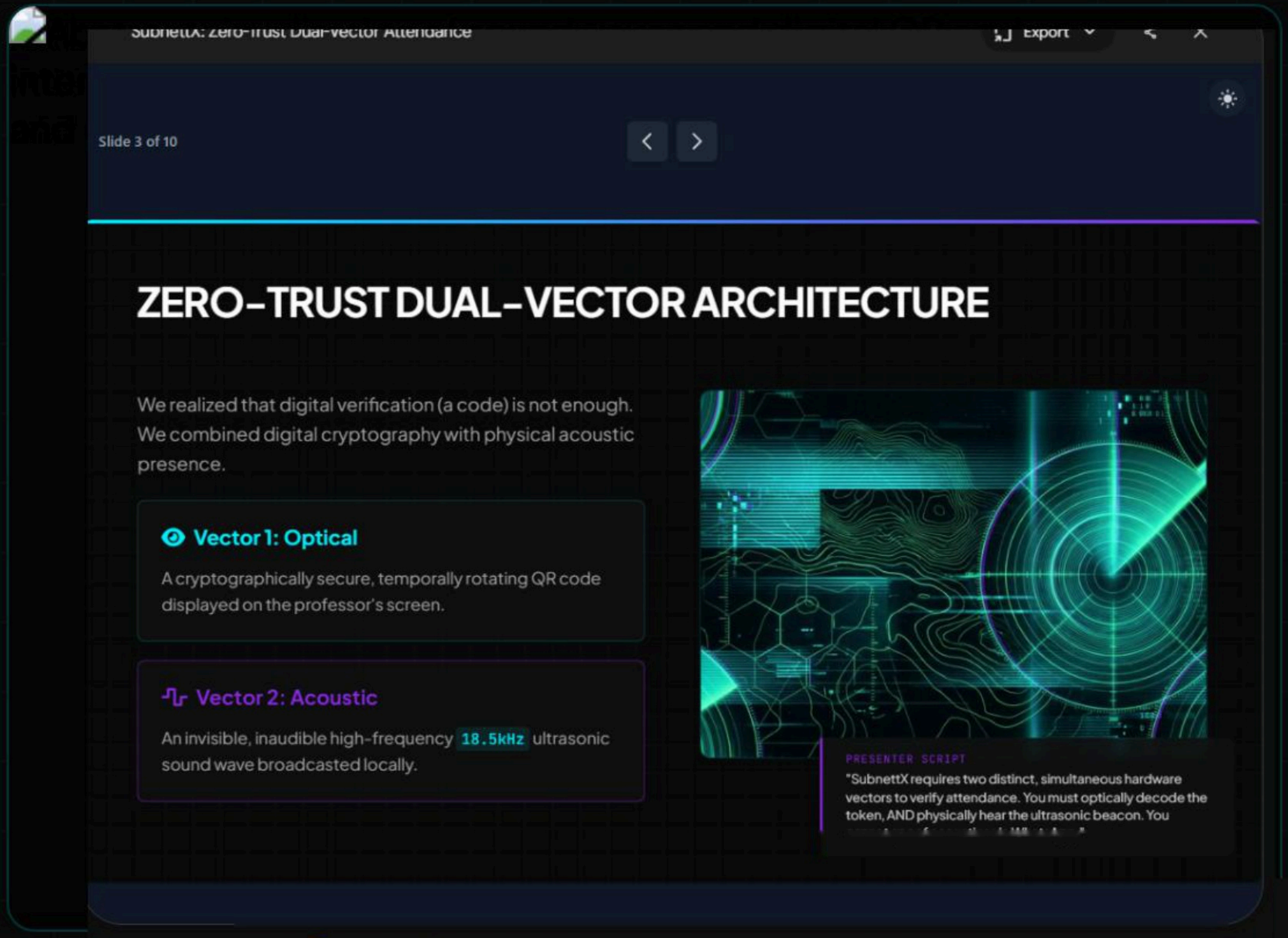
We realized that digital verification (a code) is not enough. We combined digital cryptography with physical acoustic presence.

### Vector 1: Optical

A cryptographically secure, temporally rotating QR code displayed on the professor's screen.

### Vector 2: Acoustic

An invisible, inaudible high-frequency `18.5kHz` ultrasonic sound wave broadcasted locally.



# THE 4-SECOND VERIFICATION PIPELINE

STEP_01

### The Broadcast

Professor opens the Dashboard. Laptop broadcasts a silent `18.5kHz` ultrasonic beacon & displays a rotating QR code.

STEP_02

### Optical Scan

Student points phone camera at the QR code using the SubnettX Portal.

STEP_03

### Mic Trigger

Moment the camera decodes the QR token, the phone triggers the mic via HTML5 Web Audio API.

STEP_04

### FFT Scan

Fast Fourier Transform (FFT) algorithm scans the audio spectrum for the `18.5kHz` frequency bin.

STEP_05

### Acoustic Handshake

If amplitude spikes above threshold, the acoustic handshake is locally successful

STEP_06

### Payload Delivery

System bundles Optical + Acoustic flags to

# DEEP DIVE: FAST FOURIER TRANSFORM

**18.5**

**KILOHERTZ (KHZ)**

INAUDIBLE TO HUMAN EAR

## Signal Processing on the Edge

The student's mobile browser runs real-time signal processing. It isolates the specific frequency bin to detect the professor's laptop beacon, completely ignoring standard classroom noise.

$$\text{Amplitude}_{18.5\text{kHz}} > 25 \Rightarrow \text{VERIFIED}$$

*Threshold dynamically adjusted [redacted]

# THREAT MITIGATION & BENEFITS

### Absolute Proof of Presence

If a student scans a screenshot sent by a friend, their phone's microphone will NOT hear the ultrasonic beacon in the classroom. Handshake fails instantly.

### Zero Proprietary Hardware ($0 Deployment)

Utilizes standard laptop speakers and student phone microphones. No BLE beacons or RFID scanners required for the university to purchase.

### Instantaneous Speed (SQLite WAL)

Entire handshake takes < 4 seconds. Our backend uses Write [illegible]
concurrency, allowing 50 students to hit the server simultane[illegible]

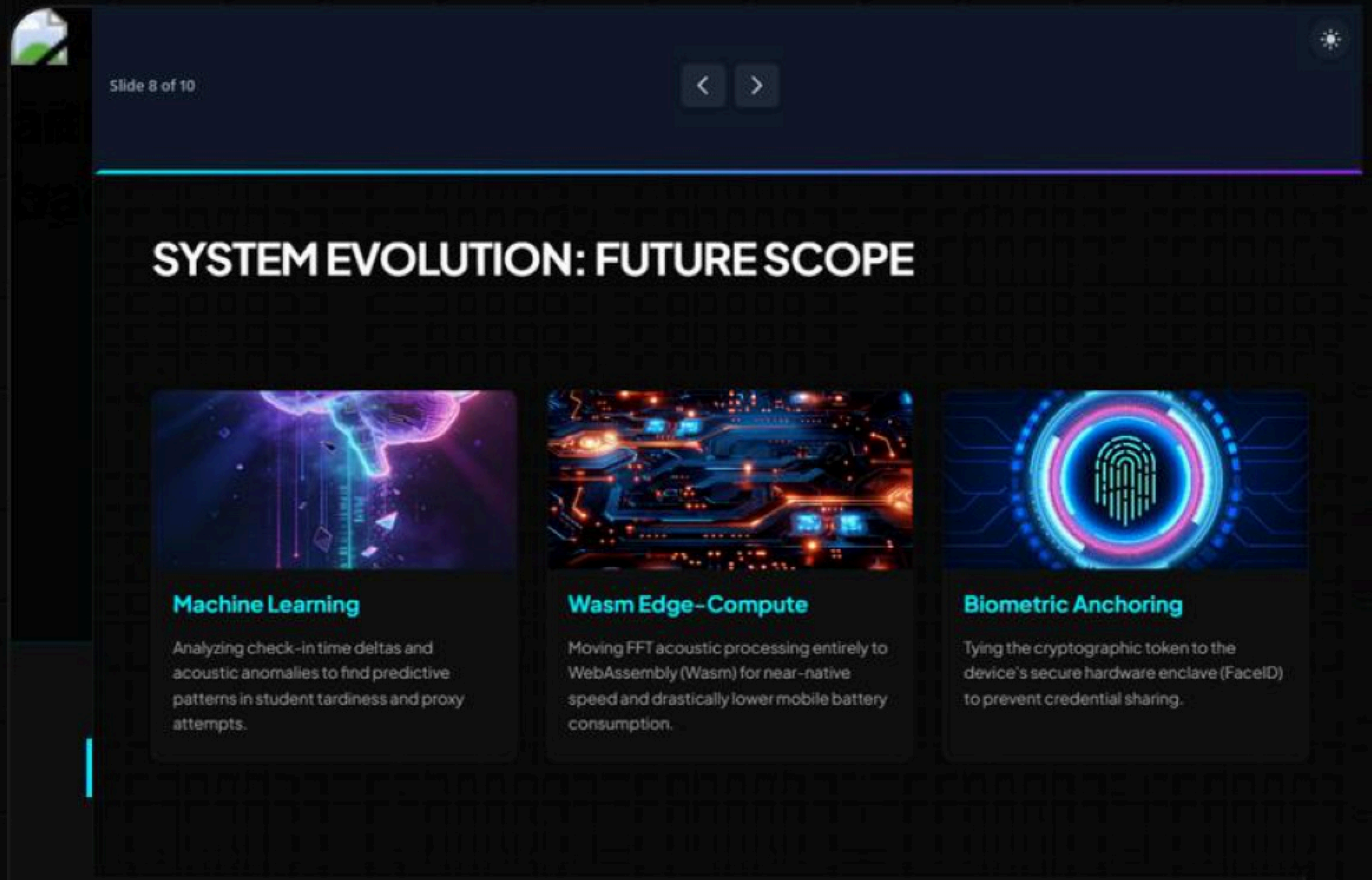
# THE THREAT DASHBOARD

We didn't just build an attendance tracker; we built a live security dashboard using long-polling updates.

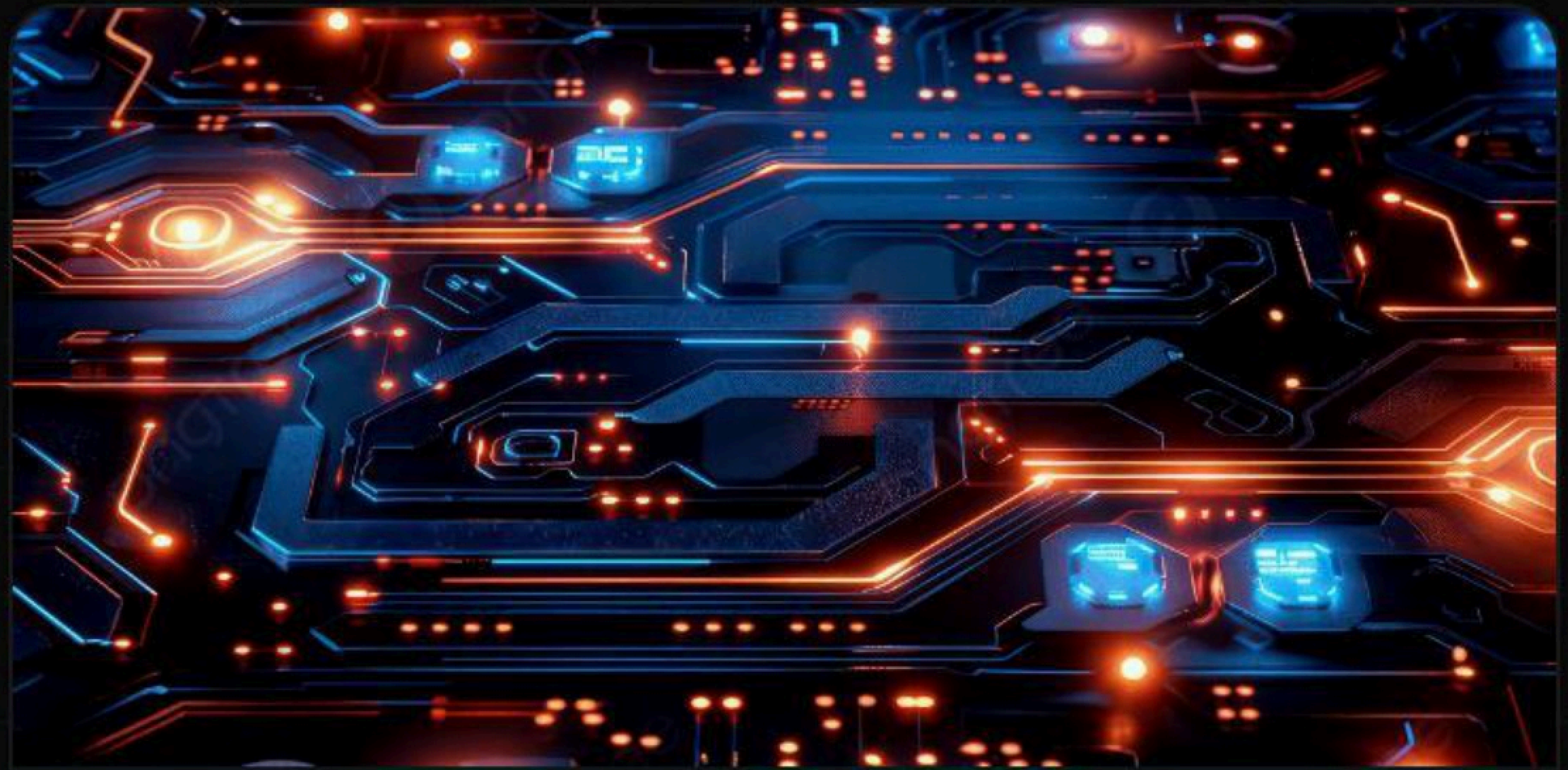
**Roll Number: CS-1042**
Optical: PASS | Acoustic: PASS

**Roll Number: CS-1088 (Spoof Attempt)**
Optical: PASS | Acoustic: FAILED

# SYSTEM EVOLUTION: FUTURE SCOPE




Analyzing check-in time deltas and acoustic anomalies to find predictive patterns in student tardiness and proxy attempts.



## Wasm Edge-Compute

Moving FFT acoustic processing entirely to WebAssembly (Wasm) for near-native speed and drastically lower mobile battery consumption.

## Biometric Anchoring

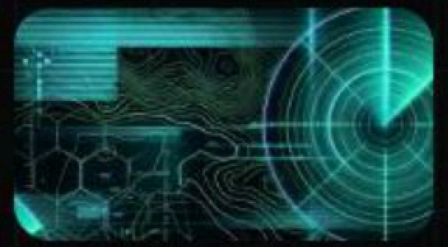
Tying the cryptographic token to the device's secure hardware enclave (FaceID) to prevent credential sharing.

# System Secured.

The future of institutional verification is Dual-Vector.

We are now open for questions.

# IMAGE SOURCES



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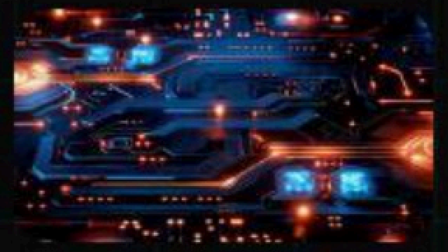
Source: stockcake.com

https://static.vecteezy.com/system/resources/previews/077/035/812/non_2x/futuristic-digital-dashboard-interface-with-data-streams-network-status-and-security-synchronization-vector.jpg

Source: www.vecteezy.com

https://images.stockcake.com/public/0/5/a/05ab8ffc-f9a5-4c70-b75f-1d6a20345894_large/neon-neural-networks-stockcake.jpg

Source: stockcake.com



https://png.pngtree.com/background/20250119/original/pngtree-a-close-up-view-of-futuristic-circuit-board-with-glowing-lights-picture-image_16064084.jpg

Source: pngtree.com

https://static.vecteezy.com/system/resources/previews/067/601/505/non_2x/futuristic-bior
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